



AI-Powered Women Safety Smart Navigation System

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SheRoute -AI



Need

- **Women's safety** during travel is a **major concern**.
- Higher **risk during night** or in unfamiliar areas.
- Existing navigation apps **prioritize shortest** or fastest route **over safety**.
- SheRoute AI recommends **safer routes** using AI.



Practical Applications

- Safe navigation for women and students or any **travelling individual**.
- Especially useful for **late-night travel** and unfamiliar locations.
- Helpful during **emergency situations** and daily commuting.



Motivation

- Improve **women's confidence** while traveling.
- Analyze **crime and environmental data** for route safety.
- **Predict safer routes** and minimize travel risks through intelligent route recommendations.
- Promote **safer** and more informed **travel decisions**.



2026

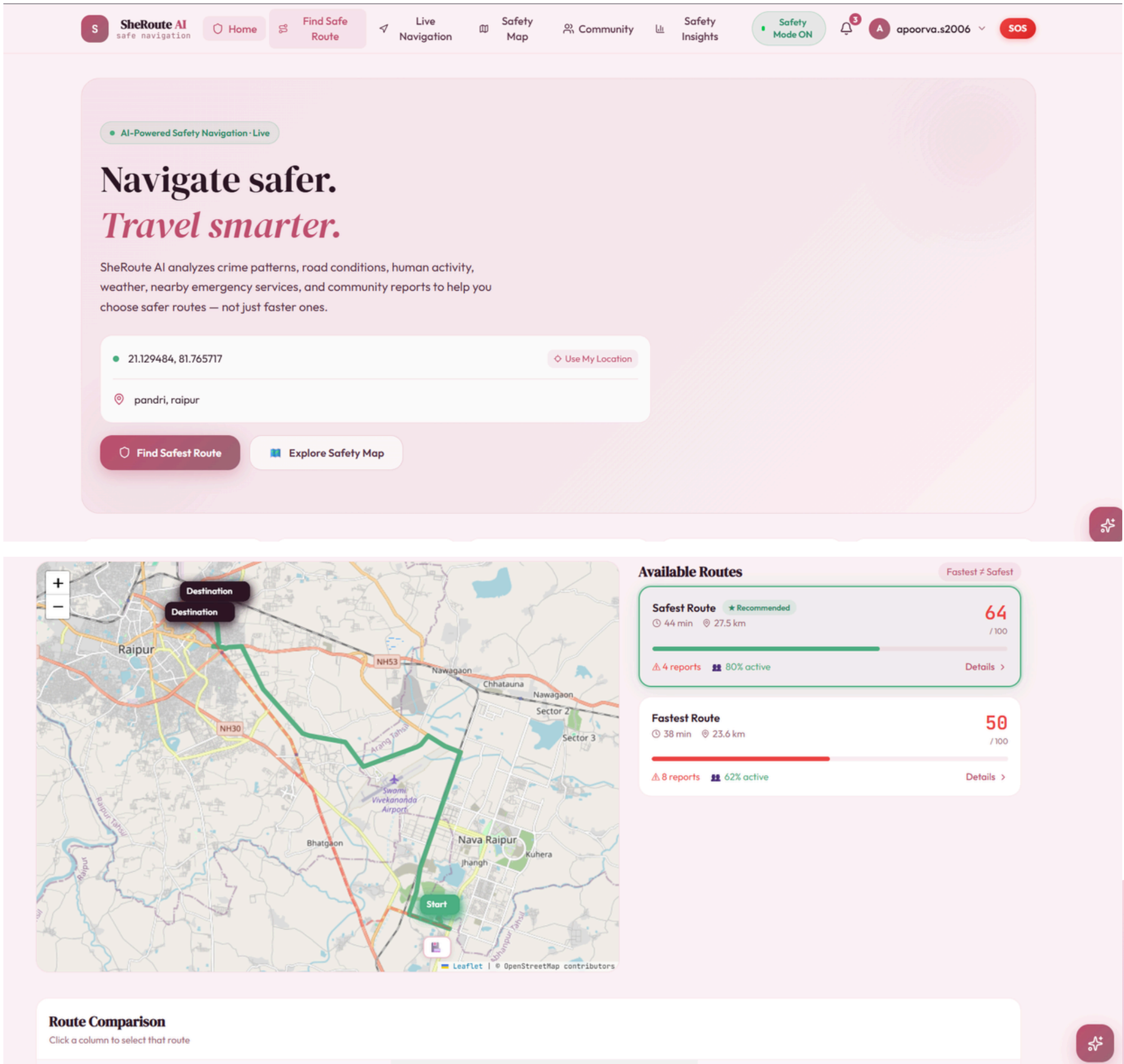


Related Works

Year	Research Paper	Main Idea	Limitation	Our Improvement
2019	GIS Safe Path Recommender	Uses GIS to find safer paths.	Uses a fixed danger index.	AI-based safety scoring with multiple safety factors.
2020	SafeRoute	Recommends safe routes using crime data.	Uses only crime history.	Added street lights, police, hospitals, community reports, and AI safety score.
2020	Women's Safety Perception (GIS)	Studies how lighting affects safety.	No route recommendation.	Provides real-time safe route recommendations.
2021	Crowd-enabled Safe Route Planning	Uses community reports for safe routes.	Depends heavily on user reports.	Uses crime data with optional community reports.
2022	Herd Routes	Uses IoT to guide users through safer crowded routes.	Requires IoT infrastructure.	Works using crime and street-light data without IoT.
2024	SafeRoutes	Uses AI and GPS for women's safety.	Limited emergency support and score explanation.	Added emergency services, risk levels, and explainable safety scores.
2025	Women Safety Platform	Predicts safe routes using machine learning.	Mainly focuses on crime data.	Combined crime data with street-light analysis.

Frontend

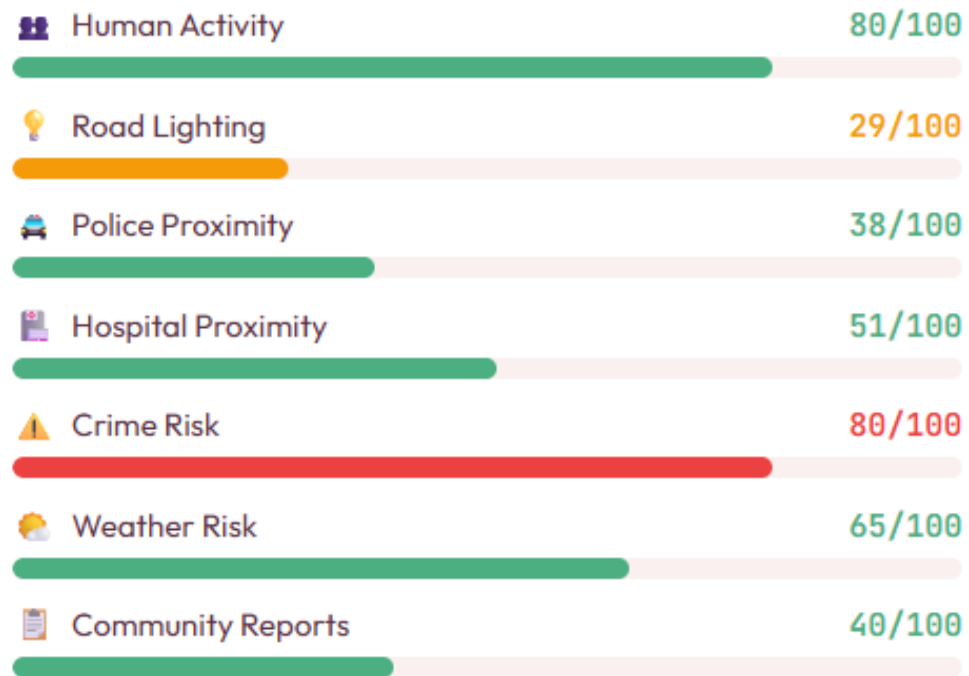
Technology	Why it is Used
React.js	Builds a fast, interactive, component-based user interface.
Vite	Provides a fast development server and optimized builds.
HTML5	Defines the structure of web pages.
Tailwind CSS	Styles the application and makes it responsive.
JavaScript	Adds interactivity and handles frontend logic.
React Router	Enables navigation between pages without reloading.
Google Maps API / Leaflet	Displays maps, routes, and user location.
Axios / Fetch API	Sends requests to your backend to get AI predictions and route data.
React Icons	Adds icons for navigation, SOS, profile, etc.





AI Safety Score

Computed from backend route analysis



↗ This route score is driven by the model output and live route factors returned by the backend.

ML prediction: **Moderate**

76.0% confidence



Why does SheRoute recommend this?

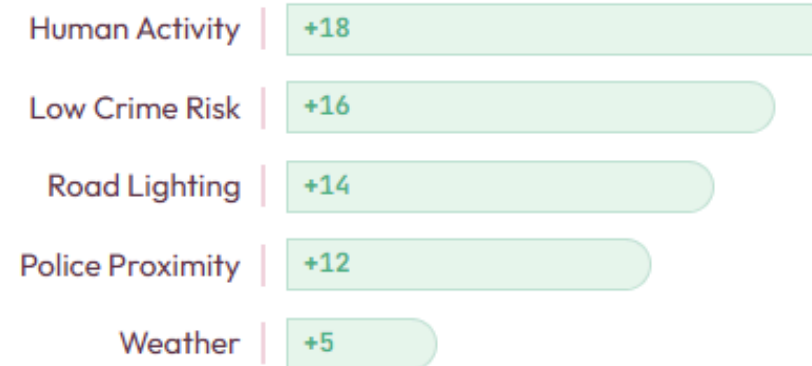
AI-powered route reasoning

76%
confident



Poor lighting detected. Prefer well-lit roads. Use caution while traveling on this route. Share your live location with a trusted contact. Keep emergency contacts easily accessible.

Safety Factor Contributions



Recent Reports

-2

Moderate Risk Route

[View Detailed Analysis →](#)



Community Reports

Safety reports from the community near your area

Moderate

🕒 2 hours ago

Broken Streetlight

Streetlight has been broken for 3 days. Very dark at night.

📍 Hill Road, Bandra West

👍 Confirm (8)

Infrastructure

Low Risk

✓ Resolved

🕒 1 day ago

Poor Road Condition

Large pothole causing traffic diversion. Poorly lit at night.

📍 Linking Road, Khar

👍 Confirm (22)

Infrastructure

Moderate

🕒 6 hours ago

Waterlogging

Heavy waterlogging after rain. Road difficult to navigate.

📍 SV Road, Santacruz

👍 Confirm (5)

Infrastructure



Live Safety Monitor

Real-time conditions along your route

LIVE

Current Safety Score

64 /100

Live panel reflects the current backend route analysis.



Weather

Rain: 40% · Vis: 7.2km

28°C · Clouds



Human Activity

Derived from the selected route profile

80%



Road Conditions

Based on route lighting score

29/100 lighting



Community Reports

Within 2km of your route

4 active



Emergency Services

Nearby services from backend

0 police · 1 hospitals



Clouds

Current route area

28°

Moderate Risk



40%

Rain



7.2km

Visibility



14km/h

Wind

Nearby Safety Services



Government Hospital Uparwara

Hospital

0.1 km

Open

Your Safety Profile



Routes Taken

47



Avg Safety Score

88/100



Reports Submitted

12



Safer Routes Chosen

38



Distance Traveled

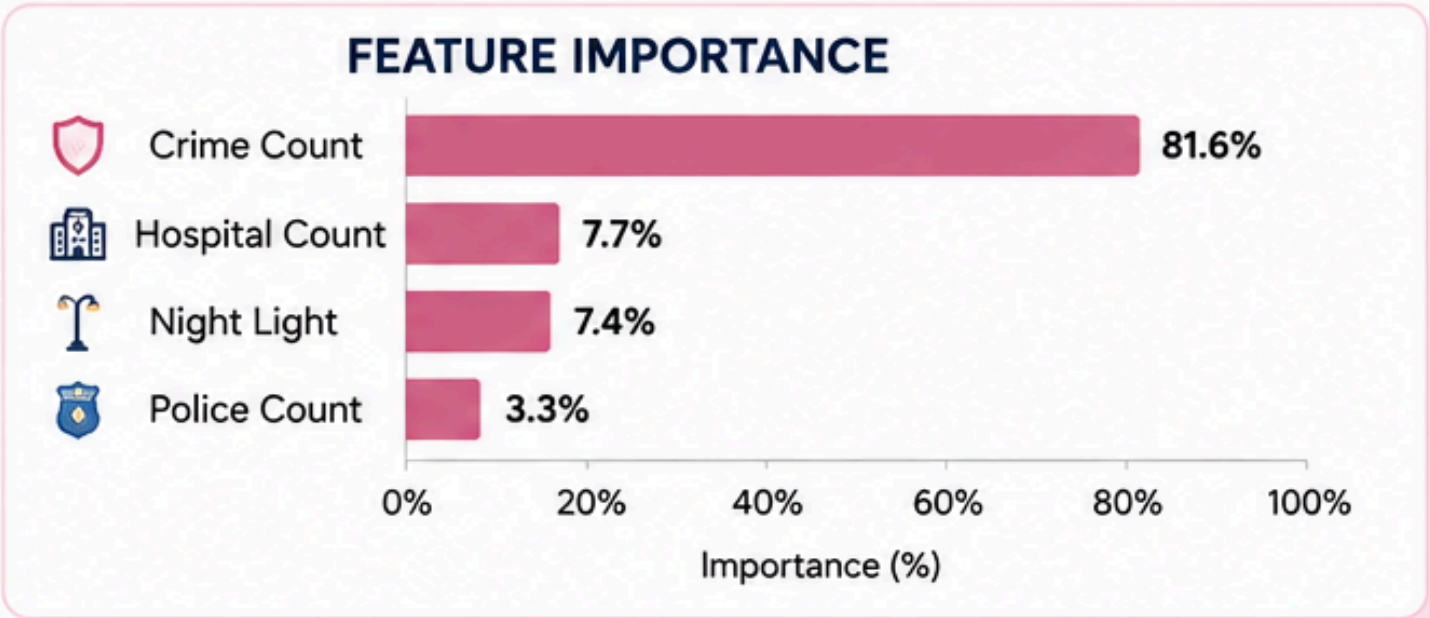
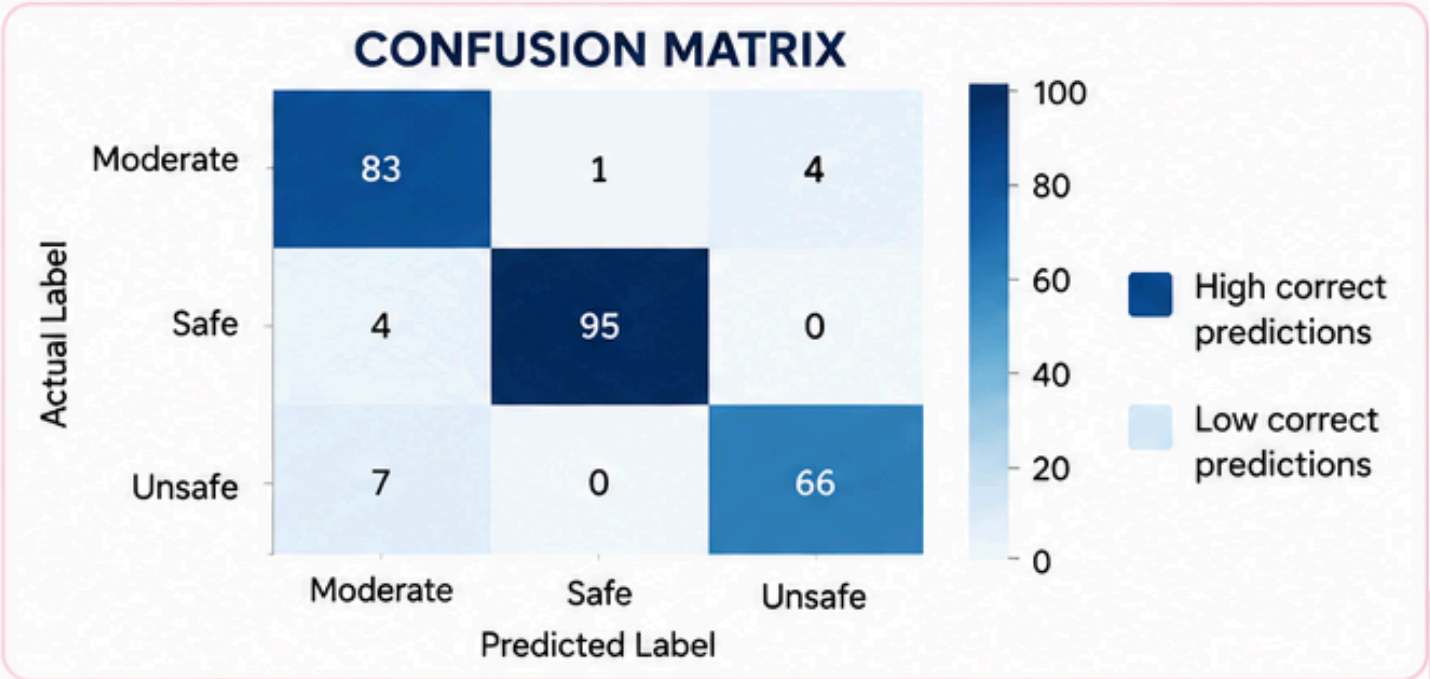
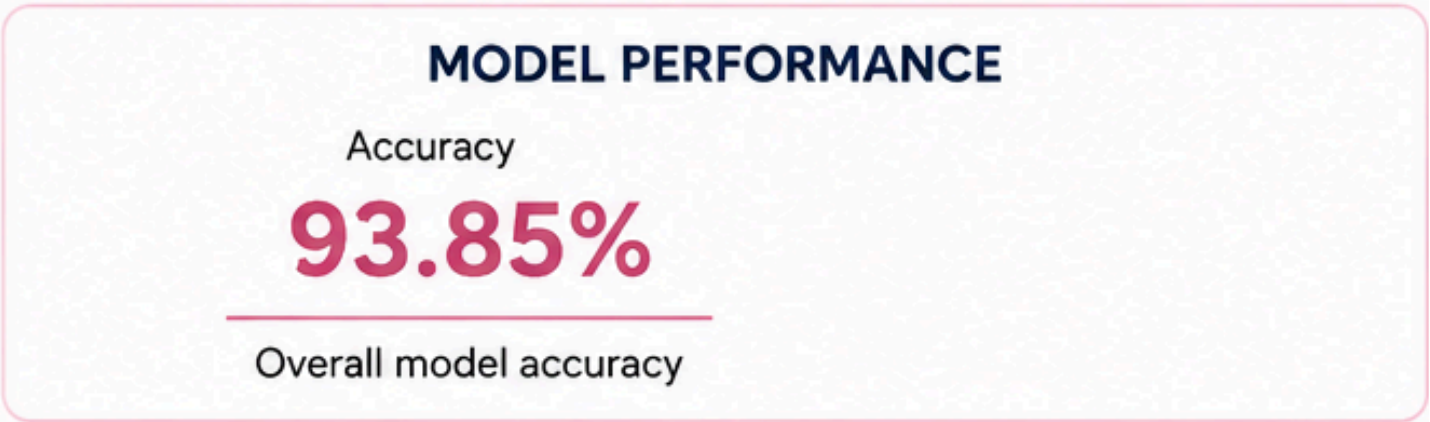
284 km



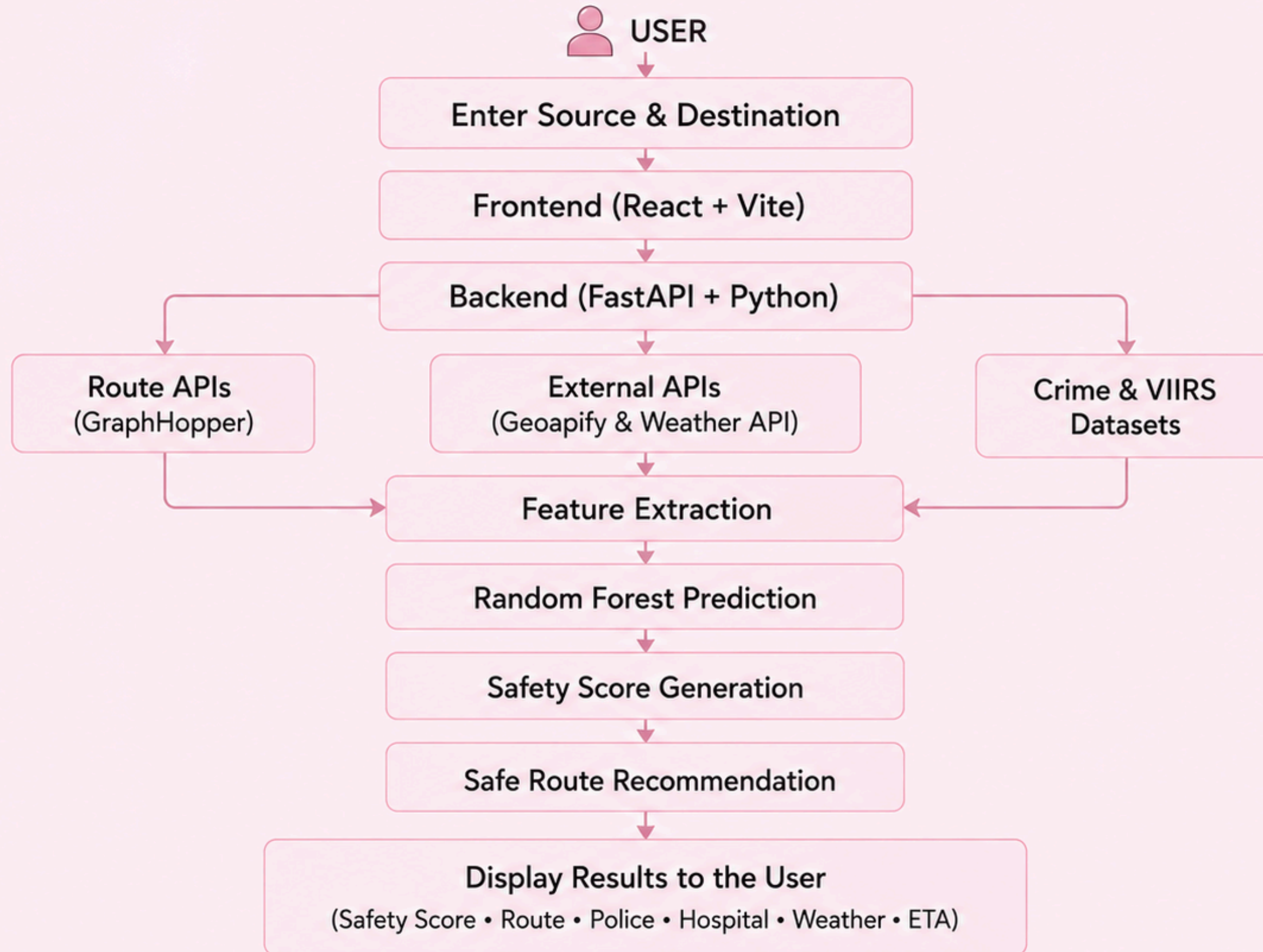
Backend

Category	Technology / Tool	Purpose
Backend Framework	FastAPI (Python)	REST API development
Database	SQLite	Data storage and management
Data Processing	Pandas, NumPy	Data manipulation and numerical computations
Route APIs	OpenRouteService (ORS), GraphHopper	Route optimization and navigation
Maps & Places API	Geoapify Places API	Location search and place information
Weather API	OpenWeatherMap	Real-time weather data
Street-Light Analysis	Rasterio (GeoTIFF)	Processing and analyzing street-light raster data
Version Control	Git & GitHub	Source code management and collaboration
API Testing	Swagger UI, Thunder Client	API documentation and endpoint testing
Backend Deployment	Render	Hosting and deployment of the backend
Frontend Deployment	Vercel	Hosting and deployment of the frontend

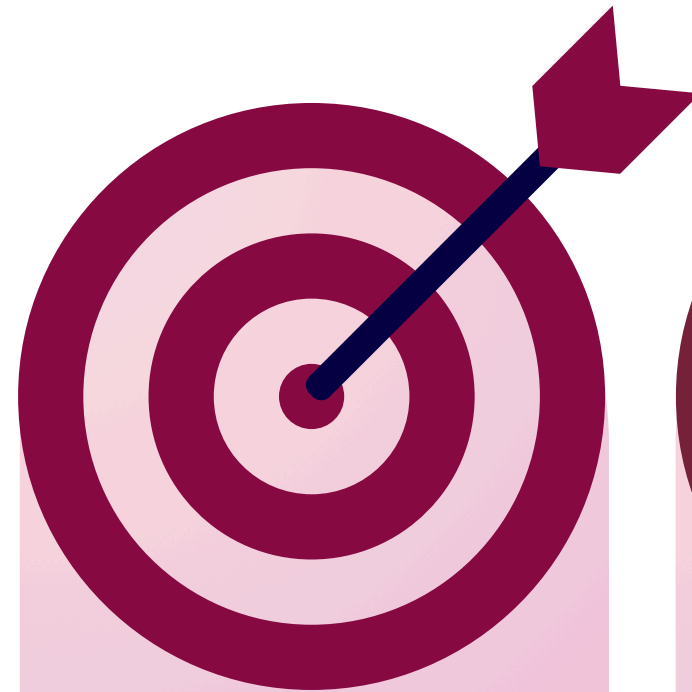
ML model		
Category		Details
Model Selection		Compared RandomForest vs XGBoost RandomForest performed better across all metrics → selected as final model
Confusion Matrix		Most confusion - Moderate and Unsafe Safe class almost never misclassified as Unsafe or vice versa
Model Validation		RandomizedSearchCV, 5-fold cross-validation, 25 configurations tested
Feature Engineering		Ratio Feartures also tested - no improvement
Conclusion		Model has extracted all learnable signal from available features
Deployment Readiness		Model serialized via joblib (sklearn-standard format) Feature extraction logic shared between training and inference (prevents train/serve mismatch)



SAFE ROUTE PIPELINE

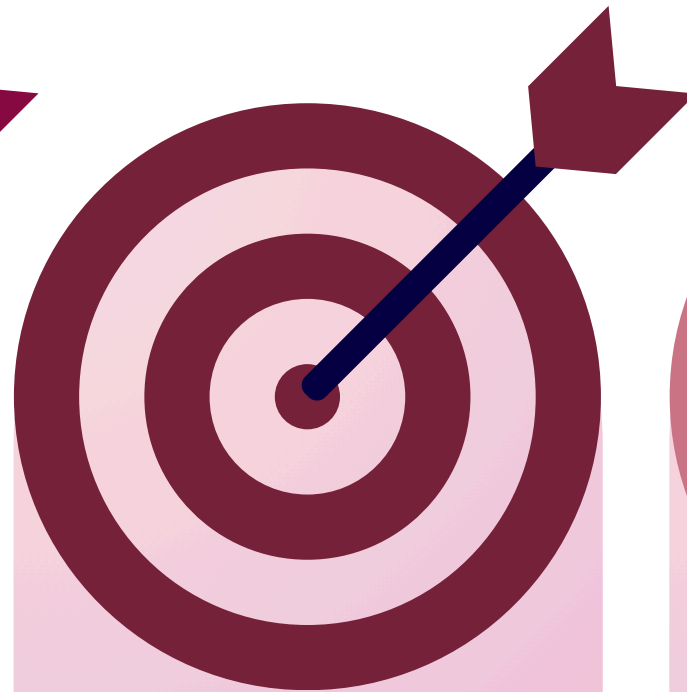


Future Work



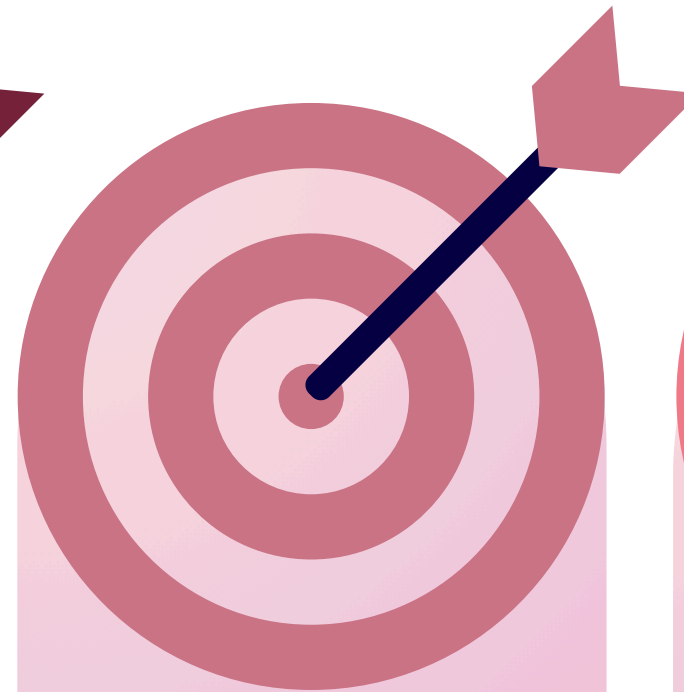
AI Crime Intelligence :

Develop advanced AI and machine learning models that analyze historical crime records, nearby safety services and human activity patterns to predict potential high-risk areas and provide more accurate safety recommendations.



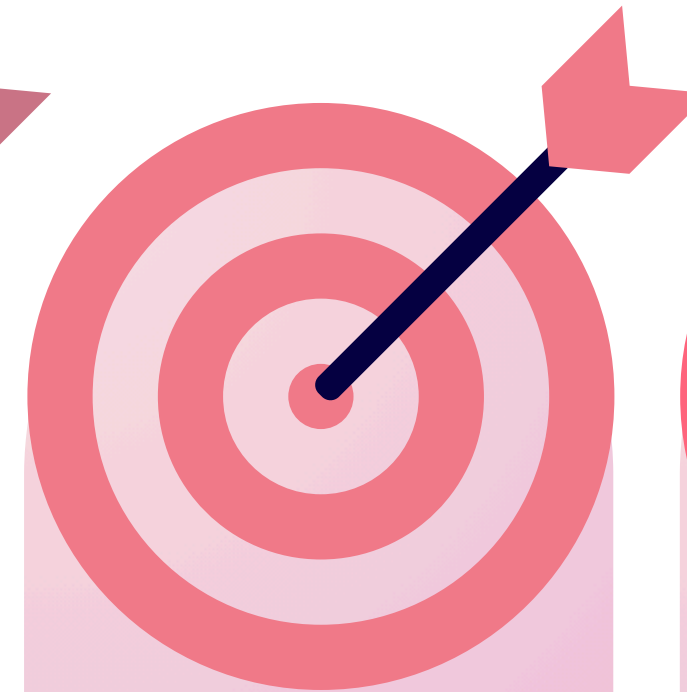
Real-time police and emergency integration:

Integrate the system with police stations and emergency response services so that SOS alerts can be shared instantly with nearby authorities, enabling faster assistance during emergencies



Voice-based emergency assistant:

Implement a hands-free voice assistant that can detect distress keywords, activate the SOS feature automatically, share the user's live location, and notify emergency contacts without requiring manual interaction.



Wearable device integration:

Connect SheRoute AI with smartwatches and fitness bands to allow quick SOS activation, continuous location tracking, and emergency alerts even when the user cannot access their mobile phone.



Smart city and predictive safety system:

Expand the platform by integrating CCTV analytics, street lighting data, traffic conditions, and community reports to generate predictive safety alerts, optimize multiple safe routes, and support deployment across multiple cities through a smart city dashboard.





THANK YOU!

WE APPRECIATE YOUR TIME AND ATTENTION.

